

CASE STUDY · AI AUTOMATION & TECHNICAL DOCUMENTATION

AI Recruiting Automation Platform

A DeepSeek matching engine and a 14-service communications pipeline, designed and built from scratch, running unattended in production.

AI Automations Expert

Senior Technical Writer

DeepSeek · LLM Workflows

Wessam Mandour

THE PROBLEM

Recruiting top-of-funnel doesn't scale by hand

Hundreds of candidates per role. Multi-channel outreach under provider rate caps. Interview transcripts to score. And a final match a recruiter must actually trust.

Volume

Hundreds of applicants per role, most not a fit, manual screening is hours of work.

Rate limits

Email + WhatsApp outreach must respect provider caps and never double-send.

Transcripts

Screening interviews arrive as raw transcripts that need consistent scoring.

Trust

A bare match score is useless, a recruiter needs to know WHY a candidate ranked.

Two systems, one automated funnel

◆ Matching Engine

Python · FastAPI · DeepSeek

Ranks candidates with a written rationale each, through a six-stage funnel. A full run takes 90 to 120 min on Railway and posts Slack status to the team.

◆ Candidate Flow

Node · Express · 14 services

2,000+ WhatsApp messages and emails a day (Twilio + Postmark), transcript scoring, and a CV parser that finishes in under a minute.

40,000

rows scanned / run

90-120 min

matching run on Railway

2,000+ /day

WhatsApp + email

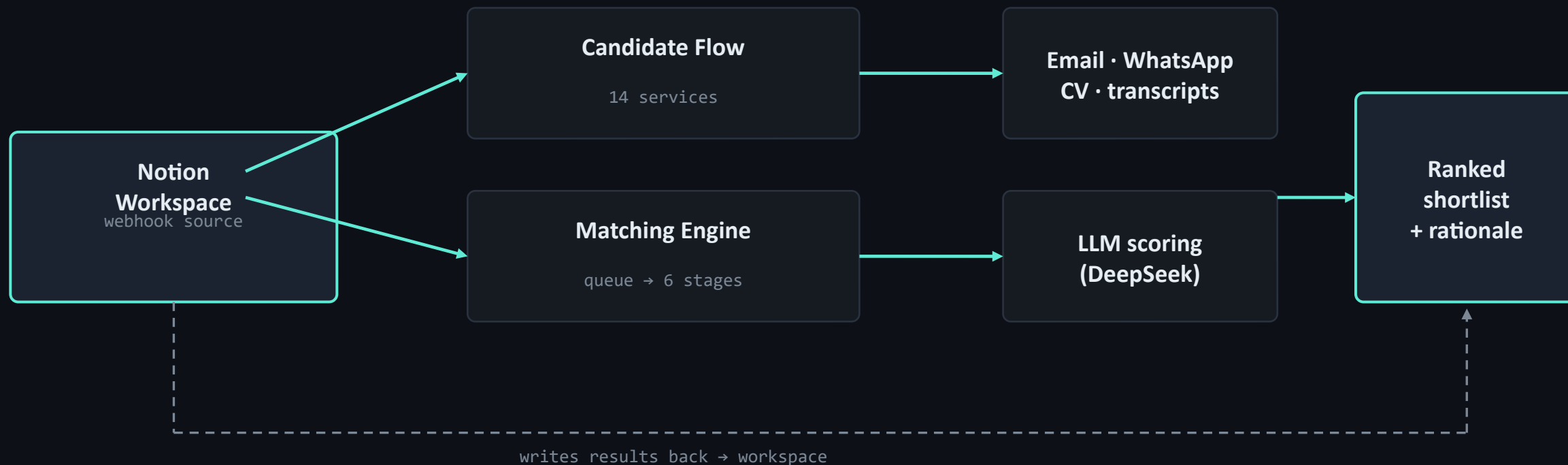
<1 min

CV parsed, regex-first

Deployed on Railway, triggered by webhooks, runs unattended, with Slack status to the whole team.

ARCHITECTURE

Event-driven, with the workspace as source of truth



No shared database to corrupt · failure in one service can't block another · every service stateless & idempotent

Six stages: job description → ranked shortlist

Each stage narrows the pool before the next, more expensive stage runs. The LLM only sees candidates worth paying to score.

1

Rubric extraction

LLM turns free-text job spec into a structured rubric: gates, requirements, keywords.

2

Keyword reduction

Cheap local filter drops no-signal candidates BEFORE any paid LLM call, top cost lever.

3

Hard filters

Non-negotiable gates: language, location, minimum screening score.

4

Priority ordering

Bucket by business priority; score the highest-value candidates first.

5

LLM scoring

0-100 + written rationale each, concurrency-capped, retried, failure-isolated.

6

Qualify & backfill

Link qualified first; backfill best below-threshold to hit target, logged.

The parts that make it production, not a script



Idempotent re-runs

Scored records skipped on re-run, no double-contact, no duplicate matches.



Rate limiting

Sequential queue, concurrency cap, write delays, under quota by design.



Graceful degradation

Backoff + JSON hardening; one bad candidate can't sink a 200-candidate run.



Schema-drift defense

Missing props ignored, odd types logged & skipped, eligibility re-checked.



AI only where needed

DeepSeek for judgment; regex for parsing, filtering, routing. No wasted tokens.



Observability + Slack

Logs keyed by id, email, stage; Slack status to the team at start and finish.

The documentation is part of the deliverable

Every service ships an operational handbook. For a system that runs unattended, the docs are the reliability layer.

Audience-layered

One-table overview for managers, branch-level depth for developers, same doc.

Operational

Documents which log line proves which behavior: debug by grep, not by reading source.

Failure-first

Maps real symptoms to the exact gate that caused them.



troubleshooting.md

```
## Troubleshooting
```

Candidate not considered for a role, check, in order:

1. Stage-2 keyword overlap
2. Stage-3 hard filters (language, location, score)
3. existing scored row (already processed)
4. duplicate-email de-duplication

Fewer qualified matches than target, expected:

Stage 6 backfills below-threshold by descending score and logs

`fallback_selected=true` per row.

What the automation delivers

40,000

candidate rows scanned per matching run

90-120 min

full run on Railway, with Slack status

2,000+/day

WhatsApp + emails (Twilio + Postmark)

<1 min

to parse a CV, regex-first, AI only when needed

Manual work that took weeks, now seconds to a couple of hours. Workflow on the previous slides.

WHAT THIS DEMONSTRATES

AI automation

14 event-driven services, idempotent, rate-limited, running unattended in production.

Technical writing

Audience-layered operational handbooks, failure-first troubleshooting, this case study.

Cost-aware AI

DeepSeek on judgment-heavy steps, regex everywhere else. No wasted tokens.

Wessam Mandour

wessam.mandour94@gmail.com · [linkedin.com/in/wessam-mandour](https://www.linkedin.com/in/wessam-mandour) · github.com/Wessam-Mandour